DIGITAL LOGIC

Chapter 3 part2:
Two Level Implementation

2025 Fall

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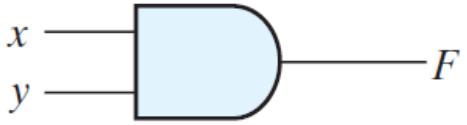
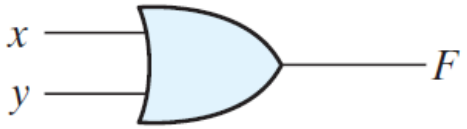
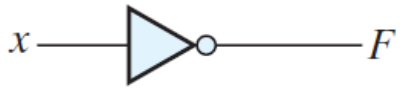
Today's Agenda

- Recap
- Context
  - NAND and NOR Implementation
  - Other Two-Level Implementations
  - Exclusive-OR Function
- Reading: Textbook, Chapter 3.6-3.9

Recall: Logic Gates

AND		$F = x \cdot y$	x y F 0 0 0 0 1 0 1 0 0 1 1 1
OR		$F = x + y$	x y F 0 0 0 0 1 1 1 0 1 1 1 1
Inverter		$F = x'$	x F 0 1 1 0
Buffer		$F = x$	x F 0 0 1 1

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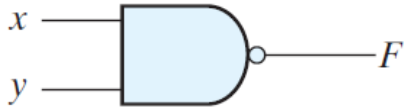
Recall: Logic Gates

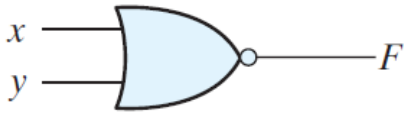
NAND



$F = (xy)'$

x	y	F
0	0	1
0	1	1
1	0	1
1	1	0

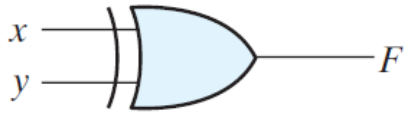
NOR



$F = (x + y)'$

x	y	F
0	0	1
0	1	0
1	0	0
1	1	0

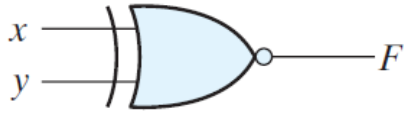
Exclusive-OR (XOR)



$F = xy' + x'y$
$= x \oplus y$

x	y	F
0	0	0
0	1	1
1	0	1
1	1	0

Exclusive-NOR or equivalence



$F = xy + x'y'$
$= (x \oplus y)'$

x	y	F
0	0	1
0	1	0
1	0	0
1	1	1

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Outline

- **NAND Implementation**
- NOR Implementation
- Exclusive-OR Function
- Other Two-Level Implementations

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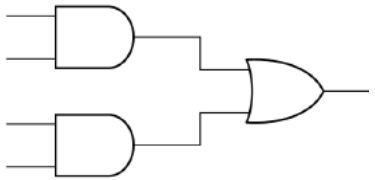
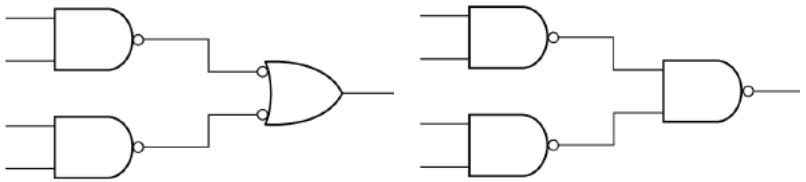
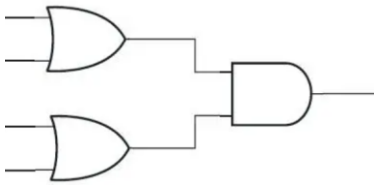
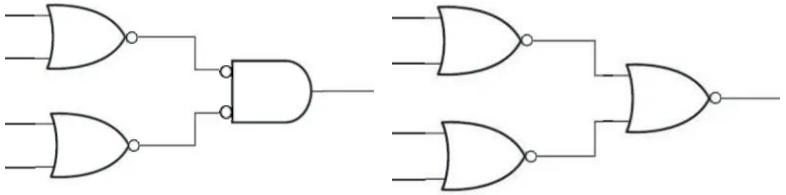
Universal Gates

- NAND gates and NOR gates are called universal gates or universal building blocks.
  - Any type of gates or logic functions can be implemented by these gates.
  - NAND and NOR gates are easier to fabricate thus are frequently used.

	Standard form	Universal Gate implementation
Sum-of-products	AND-OR 	NAND-NAND 
Product-of-sums	OR-AND 	NOR-NOR 

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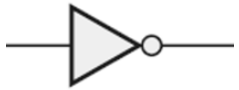
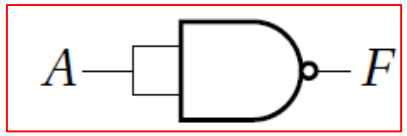
NAND circuits

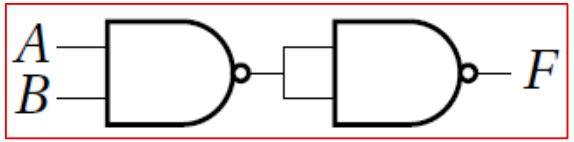
- Inverter 

  F = A' = (AA)'



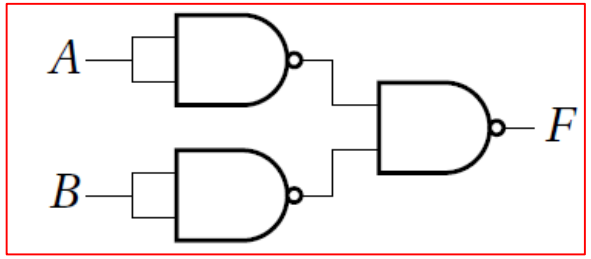
- AND 

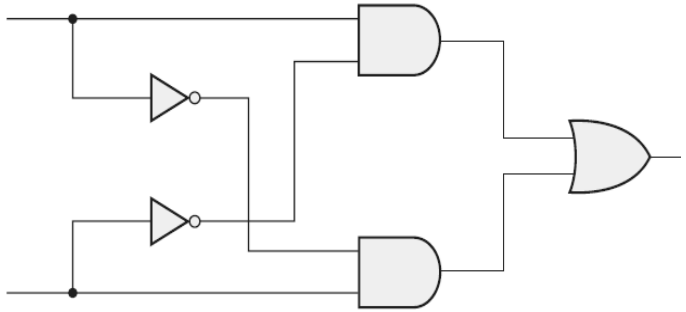
  F = AB = ((AB)')' = ((AB)' (AB)')'



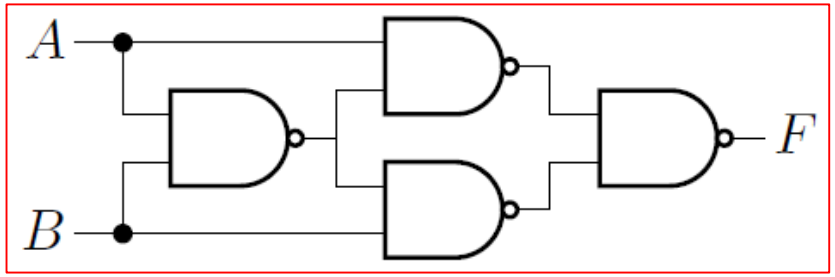
- OR 

  F=A+B=((A+B)')' = (A'B')'



- XOR 

F = AB'+A'B = AB'+A'B+AA'+BB'
= (AB'+AA') + (A'B+BB') = A(A'+B') + B(A'+B')
= ((A(AB)' + B(AB)')')' = ((A(AB)')'(B(AB)')')'



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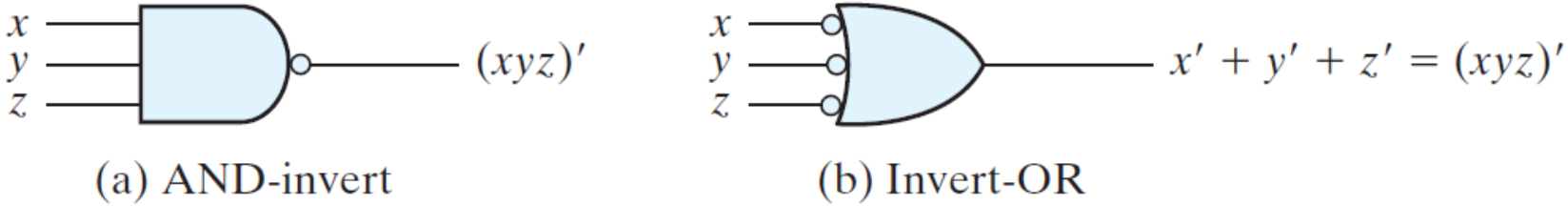
NAND circuits

- To facilitate the conversion to NAND logic, it is convenient to define an alternative graphic symbol for the gate.
- AND-invert and Invert-OR are both NAND gates



(a) AND-invert (b) Invert-OR

NAND-NAND Implementation

- A Boolean function can be implemented with two-levels of NAND gates
  1. Starting point → Simplify the function in the form of two-level **sum-of-products** (AND-OR circuit).
  2. Transfer it to two-level NAND-NAND expression.
     - algebraically (**DeMorgan's Law**)
     - or graphically (**Bubble pushing**)
  3. Draw the corresponding NAND gate implementation. A 1-input NAND gate can be replaced by an inverter.

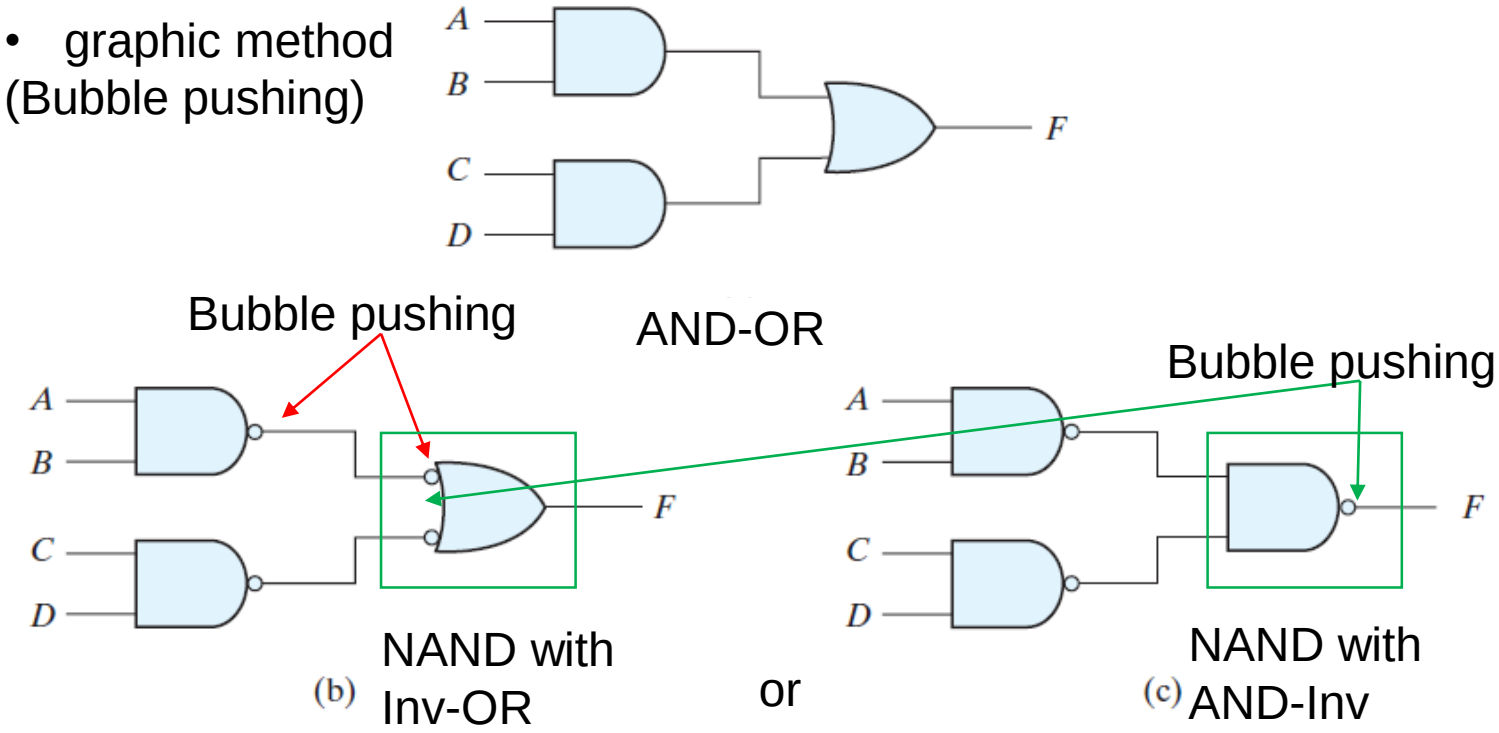
NAND-NAND Example1

- F(A,B,C,D) = AB + CD
  - Starting point: sum of products form

- algebraic method (DeMorgan's)

F = AB+CD
= ((AB+CD)')'
= ((AB)' (CD)')'

- graphic method (Bubble pushing)



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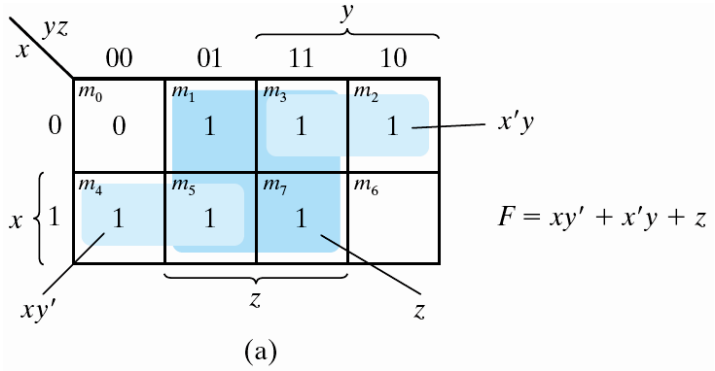
NAND-NAND Example2

- Example: Implement the following Boolean function with NAND gates
  - F(x,y,z) = ∑(1,2,3,4,5,7)
  - Starting point: K-map simplification into sum of product form

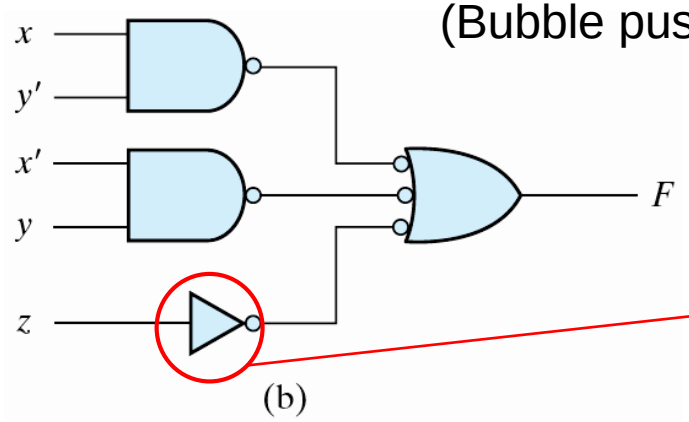


(a)

- algebraic method (DeMorgan's)

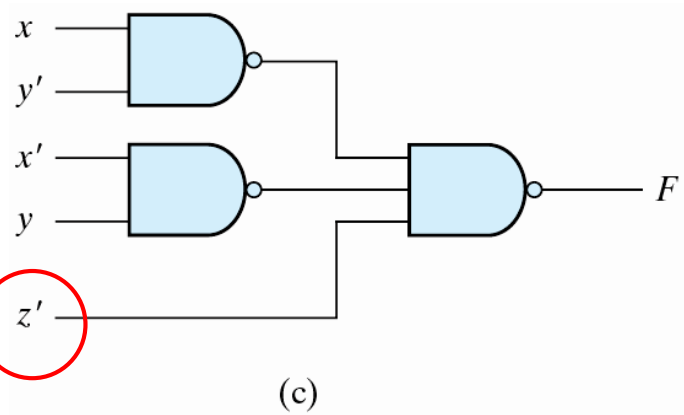
F = xy' + x'y + z
= ((xy' + x'y + z)')'
= ((xy')' (x'y)' z')'

- graphic method (Bubble pushing)



(b)

or



(c)

NAND-NAND Example3

- Exercise: Implement the following Boolean function with NAND gates
- F(A,B,C,D) = A’B’C’D+CD+AC’D

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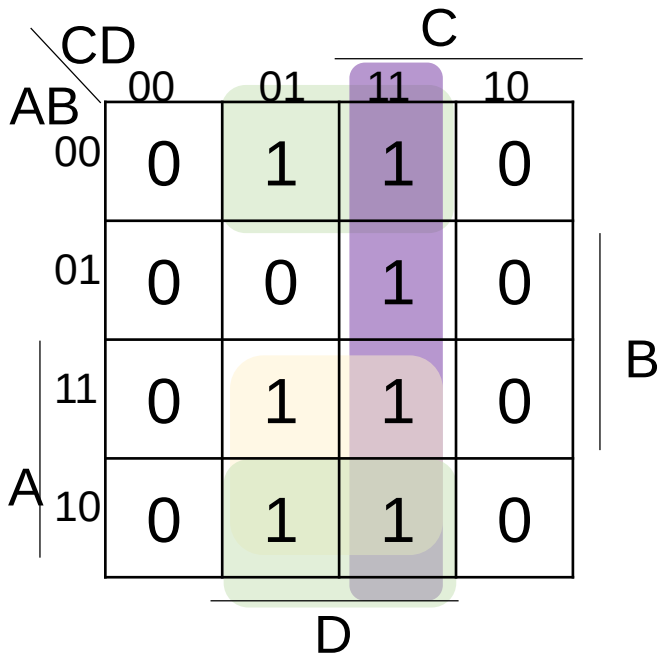
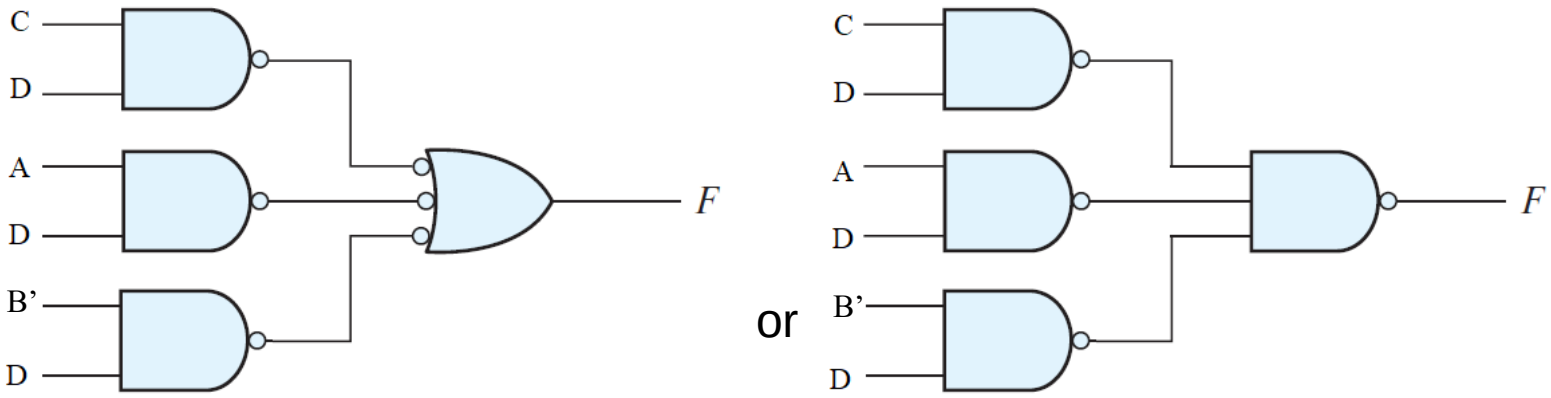
NAND-NAND Example3

- F(A,B,C,D) = A’B’C’D+CD+AC’D
  = A’B’C’D+(A+A’)(B+B’)CD+A(B+B’)C’D
  =A’B’C’D+ABCD+AB’CD+A’BCD+A’B’CD+ABC’D+AB’C’D
  =∑(1,3,7,9,11,13,15)
- Algebraic method
  F = CD+AD+B’D
  =[(CD+AD+A’D)’]’ = [(CD)’(AD)’(B’D)’]’



- Graphic method



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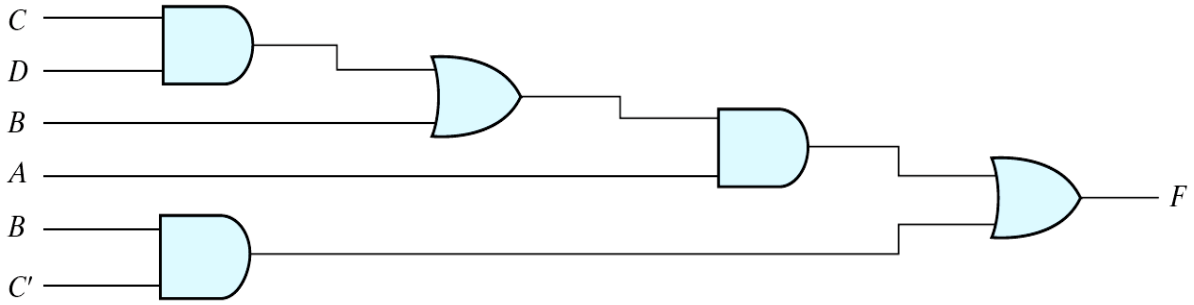
Multilevel NAND Implementation

- Multilevel-NAND circuits conversion procedure
  - Convert all AND to NAND with AND-Invert graphic symbols
  - Convert all OR to NAND with Invert-OR graphic symbols
  - Check all the bubbles (inverter) in the diagram and insert possible inverter to keep the original function

- Example: F(A,B,C,D) = A(CD+B)+BC'
  - AND-OR logic → NAND-NAND logic
    - For every bubble that is not compensated by another small circle along the same line, insert an inverter.

AND → AND + inverter

OR: inverter + OR = NAND



(a) AND–OR gates

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Multilevel NAND Implementation

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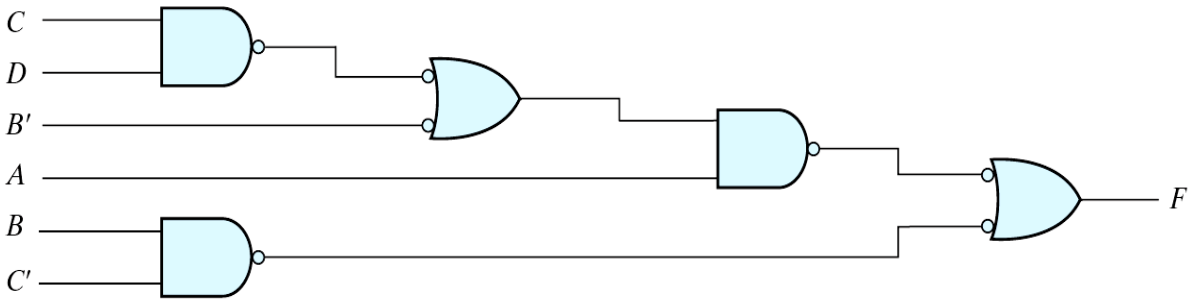
OR: inverter + OR = NAND



(b) NAND gates

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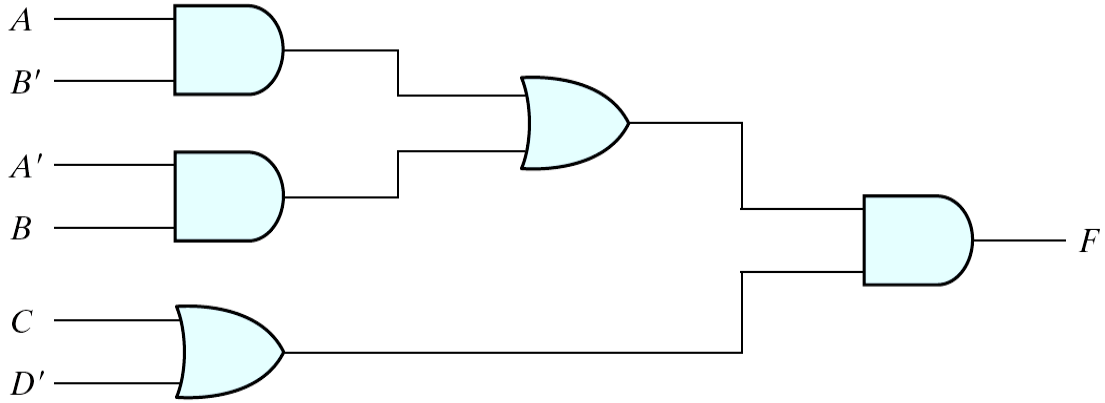
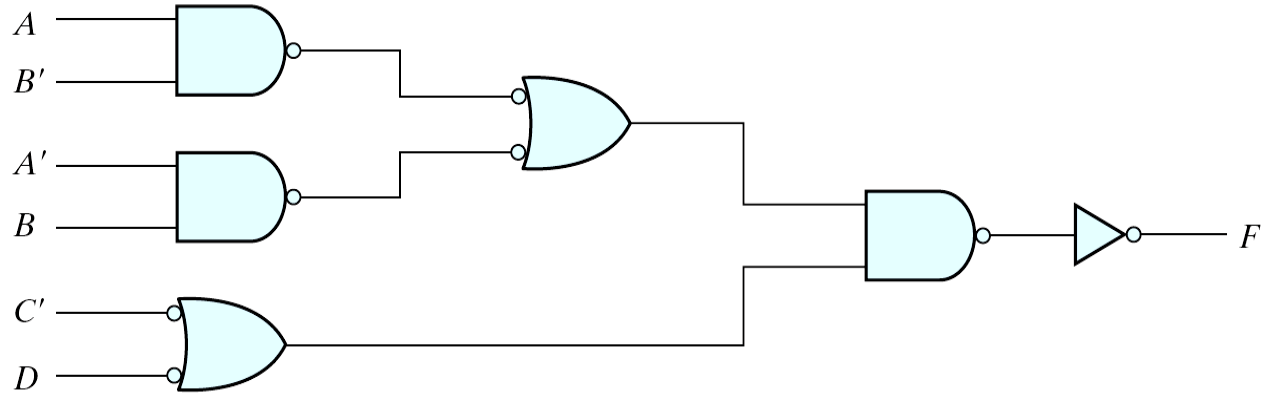
Multilevel NAND Implementation

- Exercise: Implementing F = (AB′ +A′B)(C+ D′)



(a) AND–OR gates



(b) NAND gates

Outline

- NAND Implementation
- **NOR Implementation**
- Exclusive-OR Function
- Other Two-Level Implementations

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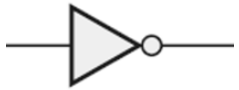
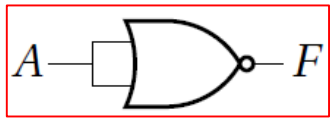
NOR circuits

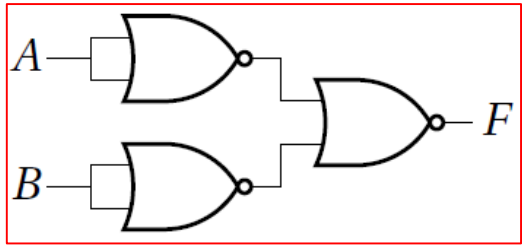
- Inverter 

  F = A’ = (A+A)’



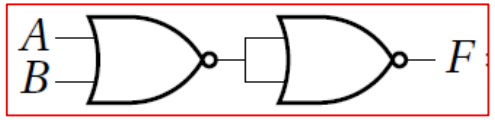
- AND 

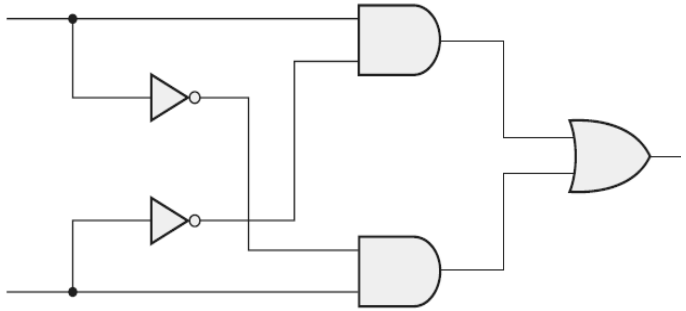
  F = AB = ((AB)’)’ = (A’+B’)’



- OR 

  F=A+B=((A+B)’)’



- XOR 

  F = AB’+A’B = …

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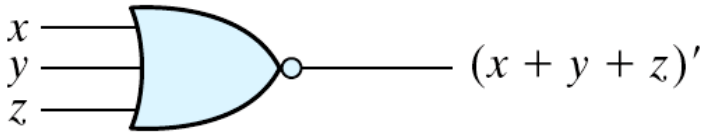
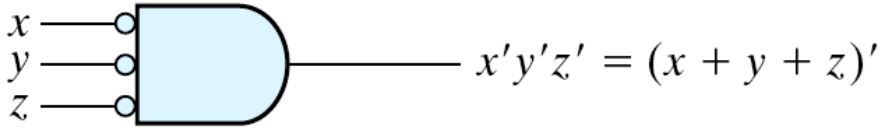
NOR circuits

- To facilitate the conversion to NOR logic, it is convenient to define an alternative graphic symbol for the gate.
- NOR-NOR is the dual of the NAND-NAND implementation
  - All procedures and rules for NOR logic are the duals of the corresponding which developed for NAND logic.



(a) OR-invert



(b) Invert-AND

NOR-NOR Implementation

- Procedure of NOR-NOR implementation
  - Starting point → Simplify the function in the form of **product-of-sum** (OR-AND circuit).
  - Transfer it to 2-level NOR-NOR expression.
    - algebraically (**DeMorgan's Law**)
    - or, graphically (**Bubble pushing**)
  - Draw the corresponding NOR gate implementation. A 1-input NOR gate can be replaced by an inverter.

sum-of-product (AND-OR) => NAND-NAND
product-of-sum (OR-AND) => NOR-NOR

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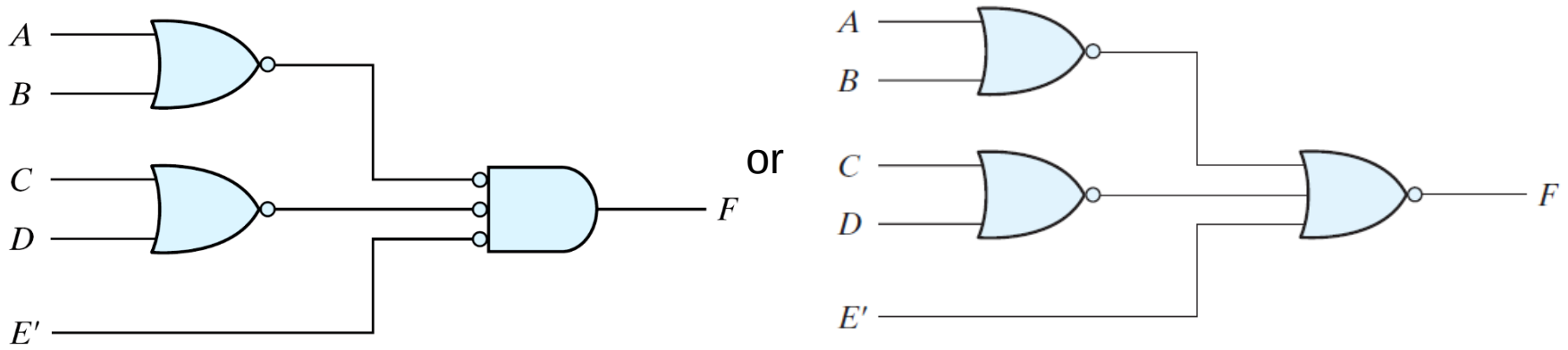
NOR-NOR Example1

- Example: Implement the following Boolean function with NOR gates
- F = (A + B)(C + D)E
  - Starting point: product of sums form → done
- Algebraic method
  - F = (A+B)(C+D)E = ((A+B)(C+D)E)')'
    = ((A+B)'+(C+D)'+E')'→DeMorgan's
- Graphic method:



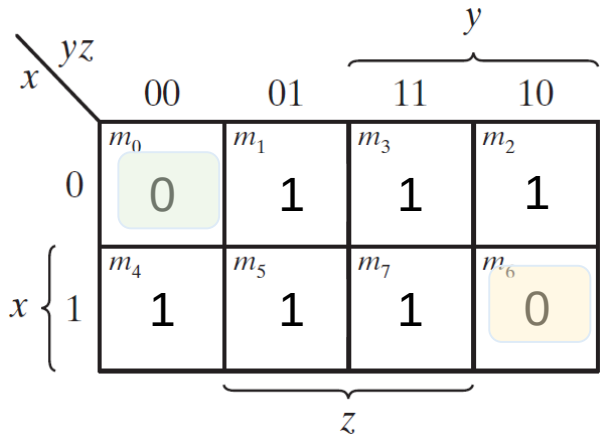
NOR-NOR Example2

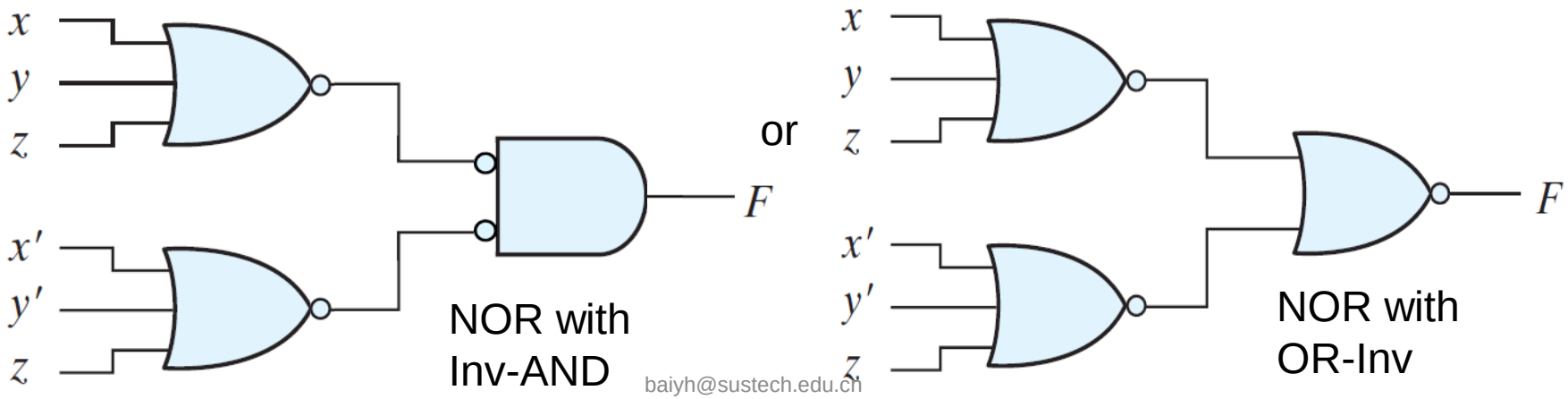
- Example F(x,y,z) = ∑(1,2,3,4,5,7)
- Algebraic method

  F'=x'y'z' + xyz'

  F = (F')' = (x'y'z' + xyz')' = (((x'y'z)')' + ((xyz')')')'

  = ((x+y+z)' + (x'+y'+z)')'
- Graphic method
  - F'=x'y'z' + xyz'
  - F = (x+y+z)(x'+y'+z) (starting point for bubble pushing)





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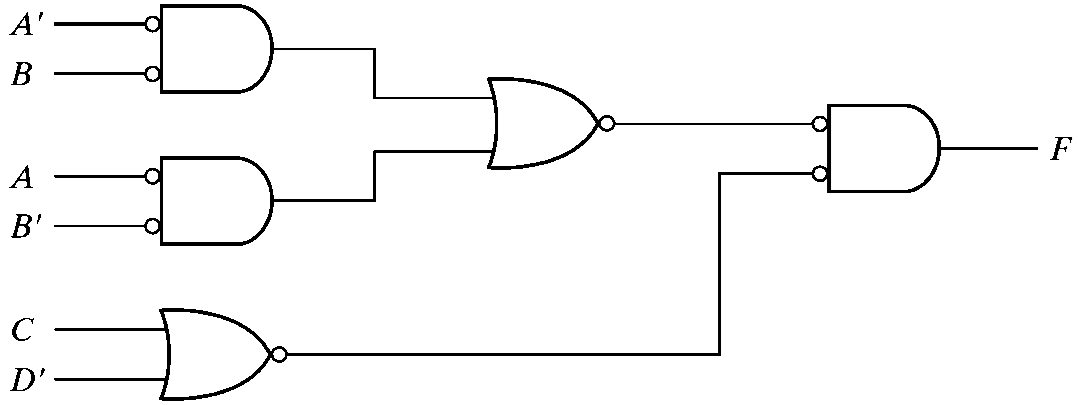
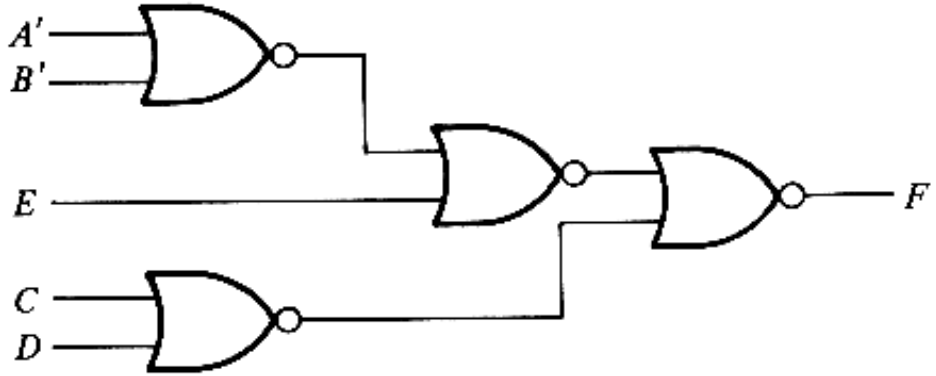
Multilevel NOR Implementation

- Change the OR gates to NOR gates with OR-invert graphic symbols and the AND gate to a NOR gate with an invert-AND graphic symbol.
- Example:
- F = (AB’ +A’B)(C + D')



- F =(AB+E)(C+D)



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Outline

- NAND Implementation
- NOR Implementation
- **Exclusive-OR Function**
- Other Two-Level Implementations

Exclusive-OR Function

- Exclusive-OR (XOR)
  - x⊕y = xy'+x'y
- Exclusive-NOR (XNOR or equivalency)
  - (x⊕y)' = xy + x'y'
- Some identities
  - x⊕0 = x
  - x⊕1 = x'
  - x⊕x = 0
  - x⊕x' = 1
  - x⊕y' = x'⊕y = (x⊕y)'
- Commutative and associative
  - A⊕B = B⊕A
  - (A⊕B) ⊕C = A⊕ (B⊕C) = A⊕B⊕C

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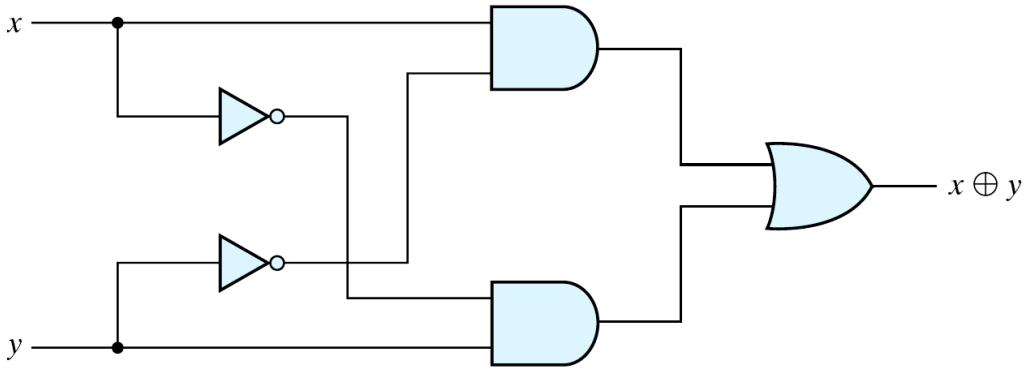
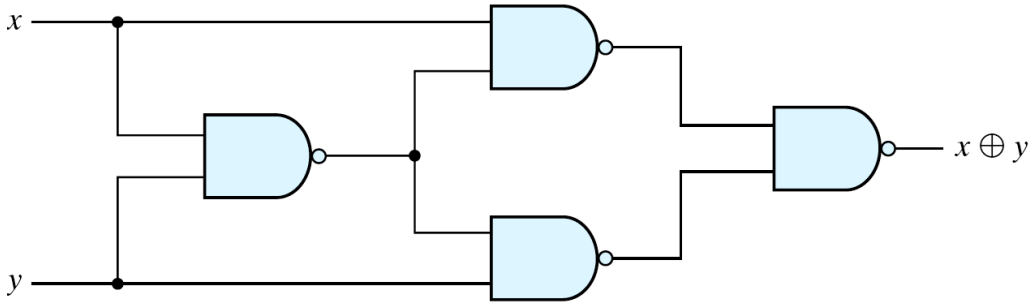
Exclusive-OR Implementations

- Implementations
  - (x'+y')x + (x'+y')y = xy'+x'y = x⊕y



(a) With AND–OR–NOT gates



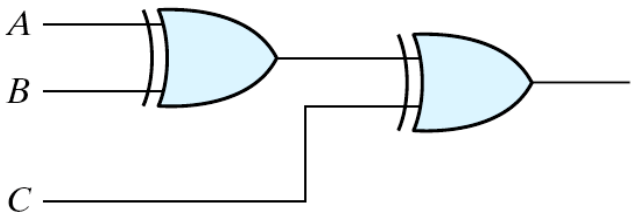
(b) With NAND gates

Odd function

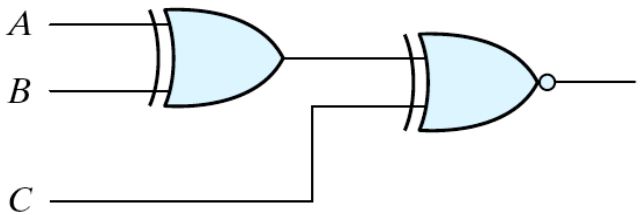
- The XOR operation with three or more variables can be converted into an ordinary Boolean function by replacing the $\oplus$ with its equivalent Boolean expression.
- $A \oplus B \oplus C = (AB' + A'B)C' + (AB + A'B')C$

  $= AB'C' + A'BC' + ABC + A'B'C$

  $= \sum(1, 2, 4, 7)$
  - Odd function (XOR) → if **odd** number of variables are equal to 1, then F = 1.
  - Even function (XNOR) → if **even** number of variables are equal to 1, then F = 1.



(a) 3-input odd function



(b) 3-input even function

Recall: Error-Detecting Code

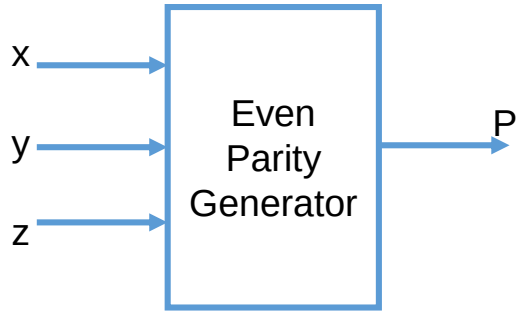
- Error-Detecting Code
  - An eighth bit is sometimes added to the ASCII character to indicate its parity.
  - A parity bit (校验位) is an extra bit included with a message to make the total number of 1's either even or odd.
- Example:

	With even parity	With odd parity
ASCII A = 1000001	01000001	11000001

Even-Parity-Generator Truth Table

Three-Bit Message			Parity Bit
x	*y*	*z*	*P*
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	0	0	0
1	1	1	1

XOR functions can be used for parity generator and parity checker



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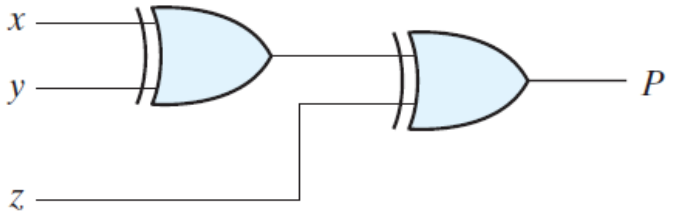
Parity Generation and Checking

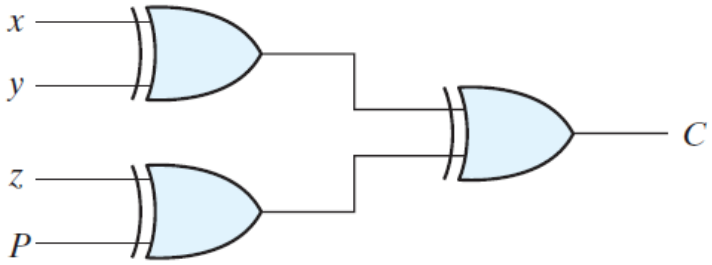
- Parity Generation circuit and Checking circuit
- Generator produces an even parity bit with: P = $x \oplus y \oplus z$
  - P = xy'z'+x'yz'+xyz+x'y'z
    = ∑(1, 2, 4, 7) – odd function



even parity generator

x	y	z	Parity bit p
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

- Checker does even parity check with: C = $x \oplus y \oplus z \oplus P$
  - C=1: one bit error or an odd number of data bit error
  - C=0: correct or an even # of data bit error



even parity checker

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Outline

- NAND Implementation
- NOR Implementation
- Exclusive-OR Function
- **Other Two-Level Implementations**

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Two-Level Forms

- AND/NAND/OR/NOR have 16 possible combinations of two-level forms
- Eight of them **degenerate** to a single operation
  - AND-AND => AND
  - OR-OR => OR
  - AND-NAND => NAND
  - OR-NOR => NOR
  - NAND-NOR =>AND
  - NOR-NAND => OR
  - NAND-OR => NAND
  - NOR-AND => NOR

Two-Level Forms

- Eight are **non-degenerate** forms
- AND-OR => standard sum-of-products
- NAND-NAND => standard sum-of-products
- OR-AND => standard product-of-sums
- NOR-NOR => standard product-of-sums
- NAND-AND/AND-NOR => AND-OR-INVERT (AOI)
  - **complement** of sum-of-products
- OR-NAND/NOR-OR => OR-AND-INVERT (OAI)
  - **complement** of product-of-sums

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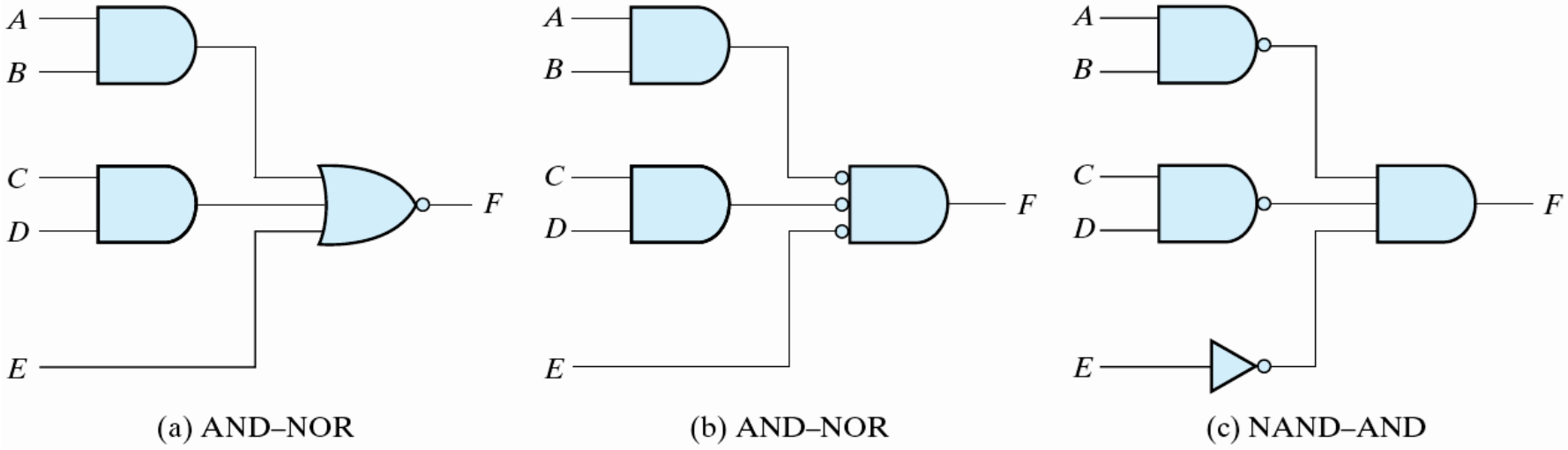
AND-OR-Invert Implementation

- NAND-AND = AND-NOR = AOI
  - F(A,B,C,D,E)=(AB+CD+E)’
  - F’(A,B,C,D,E)=AB+CD+E (sum of products)
  - An AND–OR implementation requires an expression in sum-of-products form.
  - The AND–OR–INVERT implementation is similar, except for the inversion.



Combine 0's in K-map to simplify F' in product-of-sums and then invert the results

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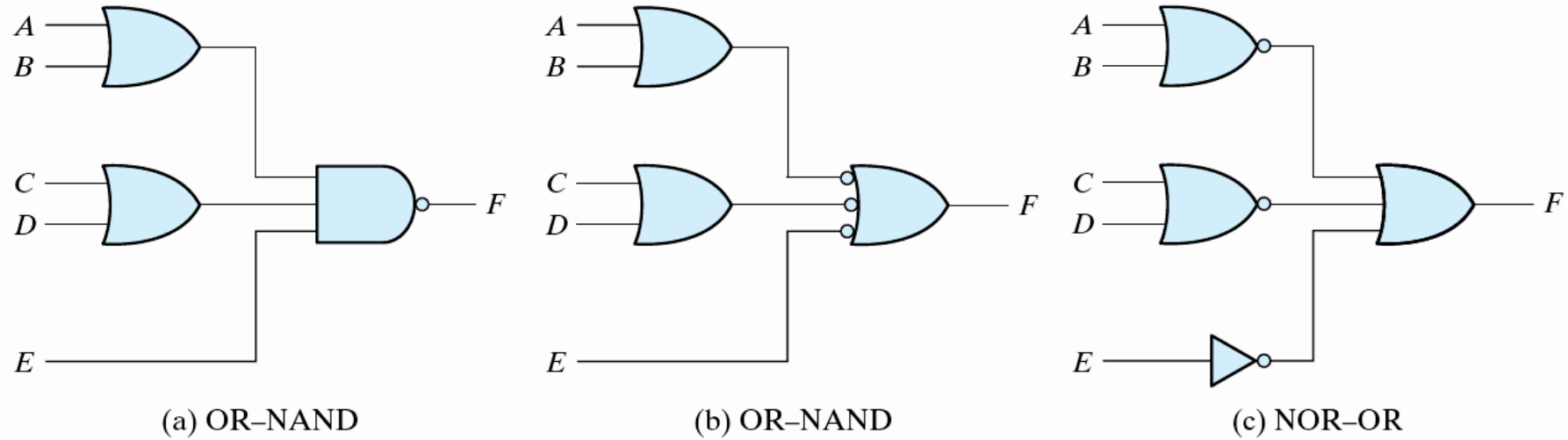
OR-AND-Invert Implementation

- OR-NAND = NOR-OR = OAI
  - F(A,B,C,D,E)=((A+B)(C+D)E)’
  - F' = (A+B)(C+D)E (product of sums)
  - The AND–OR–INVERT implementation requires an expression in product-of-sums form.



Combine 1's in K-map to simplify F' in product-of-sums and then invert the results

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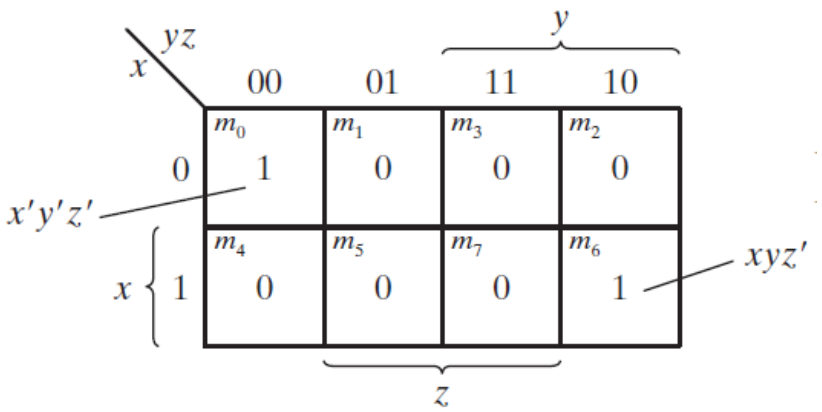
AOI & OAI Example

- Example: for the follosing circuit, find the 6 form simplified Boolean equation.



hint:
SOP for F = x’y’z’ + xyz’
SOP for F’ = x’y + xy’ + z

- AND-OR
  - F=x’y’z’ + xyz’
- NAND-NAND
  - F=((x’y’z’)’(xyz’)’)’
- OR-AND
  - F’=x’y+xy’+z → F=z'(x'+y)(x+y')
- NOR-NOR
  - F’=x’y+xy’+z → F=(z+(x'+y)'+(x+y')')'
- AOI
  - F’=x’y+xy’+z → F=(x’y+xy’+z)’
- OAI
  - F=x’y’z’+xyz’ → F’=(x+y+z)(x’+y’+z) → F=((x+y+z)(x’+y’+z))’

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Summary

- Two and multi-level implementations:
  - Universal gates: NAND and NOR gates.
  - Procedure of two-level implementations using NAND or NOR gates.
  - Procedure of multi-level implementations using NAND or NOR gates.
- Exclusive-OR gate for error detection circuits.